Unit I Assignment

1. ecommerce-app/

```
│ # React Frontend
├── client/
│   ├── src/
│   │   ├── components/
│   │   ├── pages/
│   │   ├── services/
│   │   └── App.js
│   └── package.json
│ # Express Backend
├── server/
│   ├── routes/
│   ├── controllers/
│   ├── models/
│   ├── middleware/
│   ├── server.js
│   └── package.json
│
├── database/
│   └── schema.sql
│
└── README.md
```

2. The localhost URL refers to the developer's own computer, so it will not point to the backend after deployment. An environment variable should be used instead.
   ➔ REACT_APP_API_URL=http://localhost:5000

3. .env and dotenv Code:

PORT=5000

DB_URL=mysql://root:password@localhost:3306/ecommerce

JWT_SECRET=mysecretkey

```
const express = require("express");
const dotenv = require("dotenv");

dotenv.config();

const app = express();

const PORT = process.env.PORT;
const DB_URL = process.env.DB_URL;
const JWT_SECRET = process.env.JWT_SECRET;

app.listen(PORT, () => {
    console.log(`Server running on port ${PORT}`);
});
```

4. CORS Error

npm install cors

```
const express = require("express");
const cors = require("cors");

const app = express();

app.use(cors());

app.get("/api/products", (req, res) => {
    res.json({ message: "Products data" });
});

app.listen(5000);
```

5. REST API Contract – Book Library

Endpoint	Method	Request Body	Response
/api/books	POST	{ "title": "JavaScript", "author": "John" }	Created book
/api/books	GET	None	List of books
/api/books/:id	PUT	{ "title": "React", "author": "John" }	Updated book
/api/books/:id	DELETE	None	Delete confirmation

6. Reusing Database Connection

```
const express = require("express");
const mysql = require("mysql2");

const app = express();

const db = mysql.createConnection({
    host: "localhost",
    user: "root",
    password: "",
    database: "ecommerce"
});

db.connect((err) => {
    if (err) {
        console.log("Database connection failed");
    } else {
        console.log("Database connected");
    }
});

app.get("/products", (req, res) => {
    db.query("SELECT * FROM products", (err, rows) => {
        if (err) {
            return res.status(500).send(err);
        }
        res.json(rows);
    });
```

7. package.json Scripts

➔ npm install concurrently --save-dev

```
{
  "scripts": {
    "dev": "concurrently \"nodemon server.js\" \"npm start --prefix client\""
  }
}
```

8. API Key Using Environment Variable

➔ const response = await fetch(
`https://api.weather.com/data?key=${process.env.REACT_APP_WEATHER_API_KEY}`);

env

➔ REACT_APP_WEATHER_API_KEY=your_api_key_here

10. Missing express.json() Middleware

The problem is that Express is not parsing the JSON request body, so req.body is undefined.

```
const express = require('express');
const app = express();

app.use(express.json());

app.post('/login', (req, res) => {
    const { username, password } = req.body;

    res.send(`Welcome ${username}`);
});

app.listen(3000, () => {
    console.log("Server running on port 3000");
});
```